

Total No. of Questions : 8]

SEAT No. :

PA-1499

[Total No. of Pages : 3

[5926] 119

T.E. (Information Technology)

THEORY OF COMPUTATION

(2019 Pattern) (Semester - I) (314441)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) What is a Regular Grammar? Explain types of regular grammar. [5]

b) Simplify the following CFG. [6]

$S \rightarrow ABA$

$A \rightarrow aA \mid \epsilon$

$B \rightarrow bB \mid \epsilon$

c) What is ambiguous grammar? Show that the following grammar is ambiguous and find the equivalent unambiguous grammar. [7]

$E \rightarrow E + E \mid E * E \mid (E) \mid I$

$I \rightarrow a \mid b$

OR

Q2) a) Write CFG for the language $L = \{a^i b^j c^k \mid i = j + k \text{ \& } j, k \geq 1\}$. [6]

b) Check whether the given language is CFL or not $L = \{a^n b^n c^n \mid n \geq 0\}$. [6]

c) Convert the following RLG to FA. [6]

$S \rightarrow 0A \mid 1B \mid 0 \mid 1$

$A \rightarrow 0S \mid 1B \mid 1$

$B \rightarrow 0A \mid 1S$

P.T.O.

- Q3)** a) Define Post machine. [3]
- b) Design a PDA for accepting language $L = \{ w c w^R \mid w \in (a, b)^* \}$. [6]
- c) Define Push down Automata. Explain different types of PDA. Explain any two applications of PDA. [8]

OR

- Q4)** a) Design a Pushdown Automata for the following language [7]

$$L = \{ a^n c b^n \mid n \geq 1 \}$$

- b) Convert the grammar [6]

$$S \rightarrow 0S1 \mid A$$

$$A \rightarrow 1A0 \mid S \mid \epsilon$$

to PDA that accepts the same language by empty stack.

- c) Compare Finite Automata and Pushdown Automata. [4]

- Q5)** a) Write a note on Universal Turing Machine. [5]
- b) Explain post correspondence problem with a suitable example. [6]
- c) Construct a Turing machine to find 2's complement of a binary number. [7]

OR

- Q6)** a) Design a Turing Machine to increment value of binary number by one. [8]
- b) Write short notes on [6]
- i) Unsolvable problems
- ii) Applications of Turing Machine
- c) What are recursive and recursively enumerable languages? [4]

Q7) a) What is a Traveling Salesman Problem? Justify that it is a NP-class problem. **[8]**

b) Write short notes on **[9]**

i) A Simple Un-decidable problem

ii) Measuring Complexity

OR

Q8) a) Explain Cook's theorem in detail. **[8]**

b) Explain in detail the Node-Cover Problem. **[9]**

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x

x

Total No. of Questions : 8]

SEAT No. :

P-482

[Total No. of Pages : 2

[6003]-701

T.E. (Information Technology)
THEORY OF COMPUTATION
(2019 Pattern) (Semester - I) (314441)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate marks.
- 4) Assume suitable data, if necessary.

Q1) a) Eliminate useless symbols from the following grammar. **[6]**

$S \rightarrow aA \mid bB$

$A \rightarrow aA \mid a$

$B \rightarrow bB$

b) Prove that CFL's are closed under union, concatenation, Kleene's closure. **[6]**

c) What is an ambiguous grammar? Explain with a suitable example. **[6]**

OR

Q2) a) Convert the following CFG to Chomsky Normal Form (CNF) **[8]**

$S \rightarrow aAbB$

$A \rightarrow aA$

$B \rightarrow bB \mid b$

b) Construct NFA for the following left linear regular grammar. **[6]**

$S \rightarrow B1 \mid A0 \mid C0$

$B \rightarrow B1 \mid 1$

$A \rightarrow A1 \mid B1 \mid C0 \mid 0$

$C \rightarrow A0$

c) Write a note on Pumping lemma for CFL. **[4]**

P.T.O.

- Q3)** a) Design a Pushdown Automata for the language $L = \{a^n b^{2n} \mid n > 0\}$. [6]
 b) Construct a PDA equivalent to the following CFG. [6]
 $S \rightarrow 0BB$
 $B \rightarrow 0S \mid 1S \mid 0$
 c) Write a note on Post machine. [5]

OR

- Q4)** a) Design a Pushdown Automata which accepts only odd number of b's over $\Sigma = \{a, b\}$. Simulate PDA for the string "bbaba". [8]
 b) Explain the acceptance by PDA [6]
 i) By final state
 ii) By empty stack
 c) Define Push down Automata. [3]

- Q5)** a) Design a Turing machine to accept language $L = \{a^n b^n c^n \mid n > 0\}$. [7]
 b) Write a short note on Post Correspondence problem. [5]
 c) Differentiate between Push Down Automata and Turing Machine. [6]

OR

- Q6)** a) Explain Church Turing hypothesis. [3]
 b) Design a Turing machine to add two unary numbers. [7]
 c) Write a short notes on Universal Turing machine. [8]

- Q7)** a) Explain in detail Decidable problems concerning regular languages. [5]
 b) Explain the satisfiability problem with an example. [6]
 c) What is Polynomial time reduction? Explain it with a suitable example. [6]

OR

- Q8)** a) Show that for two recursive languages L_1 and L_2 , $L_1 \cup L_2$ is also recursive. [4]
 b) What do you mean by NP-Complete problems? List the problems in this class and explain any one problem in detail. [7]
 c) What do you mean by Mapping Reducibility? Explain it with an example. [6]



Total No. of Questions : 8]

PB-3858

SEAT No. :

[Total No. of Pages : 2

[6262]-121

T.E.(Information Technology Engineering)
Theory of Computation
(2019 Pattern) (Semester-I) (314441)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) What is an ambiguous grammar? Explain with a suitable example. [4]
b) What is Regular Grammar? Explain types of regular grammar. [5]
c) Convert the following grammar to GNF. [9]

$$S \rightarrow AB$$

$$A \rightarrow BSB \mid BB \mid b$$

$$B \rightarrow aAb \mid a$$

OR

- Q2)** a) Write CFG for the language $L = \{ a^i b^j c^k \mid i = j + k \text{ \& } j, k \geq 1 \}$ [6]
b) Convert the following RLG to FA. [6]

$$S \rightarrow 0A \mid 1B \mid 0, 1$$

$$A \rightarrow 0S \mid 1B \mid 1$$

$$B \rightarrow 0A \mid 1S$$

- c) Explain any three closure properties of Context Free language. [6]
Q3) a) Define Push down Automata. Explain different types of PDA. Explain, any two applications of PDA. [6]
b) Write a note on Instantaneous Description of PDA with an example [5]

P.T.O.

- c) Convert the following CFG to PDA that accepts the same language by empty stack. [6]

$$S \rightarrow 0S1 \mid A$$

$$A \rightarrow 1A0 \mid S \mid \varepsilon$$

OR

- Q4)** a) Compare Finite Automata and Pushdown Automata. [4]

- b) Design a Pushdown Automata for the following language. [7]

$$L = \{0^a 1^b 2^c \mid a+c=b\}$$

- c) Design Post Machine for $L = \{0^n 1^n \mid n \geq 0\}$ [6]

- Q5)** a) Write a note on Universal Turing Machine [6]

- b) Explain Church Turing hypothesis. [3]

- c) Define Turing machine and design a right shifting TM over alphabet $\{0,1\}$ with an example. [9]

OR

- Q6)** a) Construct a Turing Machine to replace string '110' by '101' in a binary input string. Write down transition table along with diagram. [10]

- b) Discuss the following terms [8]

i) Post Correspondence Problem

ii) Halting Problem of Turing Machine

- Q7)** a) What do you mean by NP problems? Justify why the Travelling Salesman problem is a NP problem. [8]

- b) Define decidability of problem with suitable example. Describe un-decidable problems for context-free Grammar. [9]

OR

- Q8)** a) Write short note on : [9]

i) A Simple Un-decidable problem

ii) Measuring Complexity

- b) Explain in detail Cook's theorem. [8]



Total No. of Questions : 8]

SEAT No. :

P-7617

[Total No. of Pages : 2

[6180]-137

T.E. (Information Technology)
THEORY OF COMPUTATION
(2019 Pattern) (Semester-I) (314441)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

Q1) a) Define context free grammar. Write down CFG for the following regular expression. [6]

$(0+1)0^*(1+(01)^*)$

- b) Write a note on Chomsky hierarchy. [6]
- c) What is an ambiguous grammar? Explain with a suitable example. [6]

OR

Q2) a) What are left linear and right linear regular grammars? Explain the process of converting a Left Linear Regular Grammar to Right Linear Regular Grammar with an example. [7]

- b) Construct a CNF for the following grammar. [6]

$S \rightarrow aAbB$

$A \rightarrow aA|a$

$B \rightarrow bB|b$

- c) Check whether the given language is CFL or not $L = \{a^n b^n c^n | n \geq 0\}$ [5]

Q3) a) Construct a PDA equivalent to the following CFG. [8]

$S \rightarrow 0BB$

$B \rightarrow 0S \mid 1S \mid 0$

Test if 010^4 is in language.

- b) Compare Finite Automata and Push Down Automata. [4]
- c) Write a note on Post machine. [5]

OR

P.T.O.

- Q4)** a) Find the transition rules of PDA for accepting a language $L = \{w \in \{a,b\}^* \mid w \text{ is of } a^n b^n \text{ with } n \geq 1\}$ through both empty stack and final state and demonstrates the stack operation for the string "aaabbb". [9]
- b) Design a Post machine for the language $L = \{a^n b^n c^n\}$. [8]

- Q5)** a) Design a Turing machine to subtract two unary numbers. [6]
- b) Explain Halting Problem of turing machine. [6]
- c) Write a note on Universal turing Machine. [6]

OR

- Q6)** a) Design a Turing machine to accept language $L = \{0^n 1^n 2^n \mid n > 0\}$. Draw transition graph and write down transition table. [9]
- b) Explain the concept of Post Correspondence Problem with an example. [5]
- c) Differentiate between Finite Automata and Turing Machine. [4]

- Q7)** a) Explain in detail Decidable problems concerning regular languages, and Decidable problems concerning context free languages. [7]
- b) Explain Traveling Salesperson Problem and Justify whether the Traveling Salesperson Problem is a class P or class NP problem. [6]
- c) What are recursive and recursively enumerable languages? [4]

OR

- Q8)** a) Explain Cook's theorem in detail. [8]
- b) Explain Satisfiability Problem and comment on NP Completeness of the SAT Problem. [9]



Total No. of Questions : 8]

SEAT No. :

PC1801

[Total No. of Pages : 2

[6353]-120

T.E. (Information Technology)

THEORY OF COMPUTATION

(2019 Pattern) (Semester - I) (314441)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Convert the following CFG (Context Free Grammar) to its equivalent CNF (Chomsky Normal Form) [5]

$S \rightarrow bA|aB$

$A \rightarrow bAA|aS|a$

$B \rightarrow aBB|bS|b$

b) Explain the following terms [6]
i) Ambiguous Grammar
ii) Derivation Tree

c) Simplify the following CFG (Context Free Grammar) [7]

$S \rightarrow a|Bb|aDa|B$

$B \rightarrow D|Da| \epsilon$

$D \rightarrow b|aBb$

OR

Q2) a) Distinguish between Regular Grammar (RG) and Context Free Grammar (CFG) with example. [5]

b) Convert the following Right Linear Grammar (RLG) to its equivalent Left Linear Grammar (LLG). [7]

$S \rightarrow aA|bE$

$A \rightarrow bA|aB$

$B \rightarrow bC|aD$

$C \rightarrow bD|b|aE$

$D \rightarrow bD$

$E \rightarrow a$

c) Construct the DFA for the following Right Linear Grammar (RLG) [6]

$S \rightarrow bB$

$B \rightarrow bC|aB|b$

$C \rightarrow a$

P.T.O.

- Q3)** a) Construct the PDA for $L = \{a^n b^{m+n} c^m | n, m \geq 1\}$ [6]
 b) Construct a Post Machine to accept a language $L = \{a^n b^{n+1} | n \geq 0\}$ [6]
 c) Explain the following. [6]
 i) Deterministic PDA and Non-Deterministic PDA.
 ii) CFL for PDA

OR

- Q4)** a) Design a Pushdown Automata (PDA) for accepting the set of all strings of well formed parenthesis over inputs $(,), [,], \{, \}$ [8]
 b) Differentiate between Finite Automata and Push Down Automata. [4]
 c) Construct a Post Machine to accept the strings for a language $L = \{a^n b^n c^n | n \geq 0\}$ [6]

- Q5)** a) Design a Turing Machine for a Language $L = \{a^n b c^n | n \geq 0\}$ [7]
 b) Write a short note on [6]
 i) Recursive Language
 ii) Recursively Enumerable Language
 c) Write a short note on Multi Tape Turing Machine. [4]

OR

- Q6)** a) Design a Turing Machine to compute proper subtraction of two unary numbers. The proper subtraction function f is defined as follows. [7]

$$f(x, y) = \begin{cases} x - y & \text{if } x > y \\ 0 & \text{otherwise} \end{cases}$$

 b) Design a Turing Machine (TM) to increment a binary number by 1 [6]
 c) Write a short note on Church-Turing thesis. [4]

- Q7)** a) Write Short note on [6]
 i) Un-decidable Problem
 ii) Measuring Complexity
 b) Let $HALT_{TM} = \{ \langle M, w \rangle \mid M \text{ is a TM and } M \text{ halts on input } w \}$. Prove that $HALT_{TM}$ is undecidable. [6]
 c) Give difference between P Class and NP class Problem. [5]

OR

- Q8)** a) A_{DFA} is a decidable language. [6]
 $A_{DFA} = \{ \langle B \rangle \mid B \text{ is a DFA that accepts input string } w \}$
 b) Write a short note on Normal Forms of Boolean Expressions. [6]
 c) Write a short note on NP Completeness of SAT Problem. [5]

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Total No. of Questions : 8]

PD4322

SEAT No. :

[Total No. of Pages : 2

[6403]-120

T.E (Information Technology)
THEORY OF COMPUTATION
(2019 Pattern) (Semester - V) (314441)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Construct CFG for the language L which consists of strings over $\{0,1\}^*$ with at least one occurrence of "000". [4]
- b) Explain any three closure properties of Context Free language. [6]
- c) Convert the following grammar to CNF. [8]

$S \rightarrow PQP$

$P \rightarrow QP \mid \epsilon$

$Q \rightarrow 1Q \mid \epsilon$

OR

- Q2)** a) Consider the following grammar [8]
- $S \rightarrow aAS \mid a$
- $A \rightarrow SbA \mid SS \mid ba$
- Derive the string "aabbbaa" using
- i) Leftmost derivation
 - ii) Rightmost derivation
 - iii) Parse tree
- b) Convert the following LLRG to RLRG. [6]
- $S \rightarrow B1 \mid A0 \mid C0$
- $A \rightarrow C0 \mid A1 \mid B1 \mid 0$
- $B \rightarrow B1 \mid 1$
- $C \rightarrow A0$
- c) Explain Chomsky Hierarchy of grammar. [4]

P.T.O.

- Q3)** a) Construct PDA for the following CFG. [6]
 $S \rightarrow 0SX \mid 1SY \mid \epsilon$ $X \rightarrow 1$ $Y \rightarrow 0$
 b) Write formal definitions of [4]
 i) Deterministic PDA
 ii) Non-deterministic PDA
 c) Define post machine. Design PM to check well formedness of parentheses. [7]

OR

- Q4)** a) Compare finite automata and push down automata. [4]
 b) Design a PDA for the language $L = \{a^n b^m c^n \mid m, n \geq 1\}$ [6]
 c) Define Push down Automata. Explain different types of PDA. Explain any two applications of PDA. [7]

- Q5)** a) Explain the power of Turing machine over Finite Automata. [4]
 b) Design a Turing machine to accept language $L = \{0^n 1^n \mid n \geq 1\}$. Explain with diagram and transition table. [8]
 c) Explain Halting Problem of Turing Machine. [6]

OR

- Q6)** a) Write a note on variants of Turing machine. [8]
 b) Explain solvable problems and unsolvable problems. [4]
 c) Construct a Turing machine for 1's complement of a binary number. [6]

- Q7)** a) Show that for two recursive languages L_1 and L_2 , each of the following are recursive. [9]
 i) $L_1 \cup L_2$
 ii) $L_1 \cap L_2$
 b) Write a note on Node-Cover Problem. [8]

OR

- Q8)** a) What do you mean by polynomial time reduction? Explain with a suitable example. [8]
 b) Define P-class problems and NP- class problems. Explain them with suitable examples. [9]

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